

LANET: Language Network

Feature Expansion & Development Document

AI-Powered Ethiopian Language Learning Platform

Project Overview

This document outlines the new feature expansion plan for LANET, a modern AI-powered Ethiopian language learning platform. The features focus on improving conversation skills, listening, reading, writing, social engagement, AI innovation, and progress analytics. These enhancements aim to provide learners with a comprehensive, interactive, and personalized language learning experience.

1. Conversation Practice

- Role-play scenarios (ordering food, job interviews, travel, daily conversations).
- Voice-based conversations .
- Pronunciation feedback.

2. Listening Skills

- Audio lessons organized by proficiency level.
- Interactive listening quizzes.

3. Reading Features

- Short stories with clickable vocabulary definitions.
- News articles adapted by language level.
- Dual-language reading mode (native and target language).

4. Writing Features

- Sentence construction exercises.
- Open-ended writing prompts.
- Personalized vocabulary lists.
- Weakness tracking dashboard.

5. Social & Community Features

- Community discussion boards.
- Correction exchange between learners.
- Group challenges and competitions.

6. Advanced / Innovative Features

- Object labeling.
- Real-world vocabulary detection.

7. Progress & Analytics

- Weekly performance reports.
- Skill-based progress tracking (Listening, Speaking, Reading, Writing).
- Personalized improvement recommendations.
- Learning streak tracking.

Project Impact

These new features will transform LANET into a comprehensive language learning ecosystem. The platform will support Ethiopian languages, improve communication, preserve culture, and provide modern AI-powered education tools.

HOSPITAL MANAGEMENT SYSTEM (HMS)

Digital Healthcare Management Platform

Introduction

Healthcare facilities in Ethiopia and many developing regions face major challenges in managing patient data, appointments, and hospital operations efficiently. The Hospital Management System provides a modern, digital solution that streamlines healthcare services, improves patient care, and enhances operational efficiency.

Why Current Systems Fall Short:

- No centralized patient record management
- Paper-based systems causing data loss and delays
- Poor appointment and scheduling management
- Limited access to patient history and reports

The Challenge

- Healthcare management inefficiencies affect multiple critical areas:
- Patient Management
Difficulty tracking patient history, treatments, and records
- Appointment Scheduling
Long waiting times and poor scheduling coordination
- Treatment & Medical Records
Incomplete or lost medical records affecting care quality
- Hospital Administration
Poor reporting, billing, and operational management

Our Solution

The Hospital Management System transforms healthcare operations through digital automation, secure data management, and intelligent workflows.

Key Features

- Patient Management
Digital patient registration and medical history tracking
- Appointment Scheduling
Easy booking, rescheduling, and appointment management
- Doctor Management
Manage doctors, schedules, and patient assignments
- Medical Records
Secure storage of diagnoses, prescriptions, and reports

- **Billing & Payments**
Automated billing and payment tracking
- **Reports & Analytics**
Generate operational and medical reports
- **Secure & Reliable**
Role-based access and secure patient data
- **Multi-Platform Access**
Accessible via web and mobile devices

Technology Stack

Frontend:

Web: React.js / Next.js + TypeScript

Backend:

Node.js / Go (Golang)

Database:

PostgreSQL

Infrastructure:

Docker

REST API

Project Impact

- Improves quality of healthcare services
- Reduces paperwork and administrative workload
- Provides fast and secure access to patient records
- Improves hospital efficiency and decision-making
- Supports digital transformation of healthcare

Hospital Management System

Digitalizing Healthcare for a Better Future

SHOPNET: SMART E- COMMERCE PLATFORM

Online Shopping and Business Management System

Introduction

E-commerce is transforming how businesses and customers interact. ShopNet provides a modern, scalable, and intelligent online shopping platform that enables businesses to sell products online while offering customers a seamless shopping experience.

Why Current Solutions Fall Short:

- Limited access for small and medium businesses
- Poor user experience and slow performance
- Lack of intelligent product recommendations
- Limited mobile and offline capabilities

The Challenge

Businesses and customers face multiple challenges in online commerce:

Customer Experience

Difficulty finding products and slow checkout processes

Business Management

Manual inventory tracking and inefficient sales management

Order Processing

Lack of real-time order tracking and management

Data & Insights

Limited analytics for business growth and decision making

Our Solution

ShopNet revolutionizes online commerce through modern technology.

Key Features

Smart Shopping Cart

Easy product management and fast checkout

Order Management System

Real-time order tracking and status updates

Admin Dashboard

Manage products, users, and sales analytics

Secure Payments

Safe and reliable payment integration

Technology Stack

Frontend:

Web: Next.js + TypeScript

Backend:

Node.js + Express or Go (Gin)

PostgreSQL or MongoDB Database

Other Tools:

Docker

Cloud Hosting (AWS, Azure, or Vercel)

Project Impact

- Enables businesses to sell products online easily
- Improves customer shopping experience
- Automates business operations

- Supports digital transformation
- Creates new business opportunities

ShopNet - Empowering Digital Commerce